



Optimization strategies of **Bayesian global optimization** Platform – Bgolearn | MLMD

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If you use the PPT, please quote it as follows :

Bin CAO. (2024). Bgolearn: A Bayesian global optimization package. Retrieved from <https://github.com/Bin-Cao/Bgolearn>



Self introduction



曹斌 (Bin CAO)

- 🎓 2023.9-Present : Hong Kong University of Science and Technology(GZ) / **PHD student** / Supervisor : Prof. Zhang Tongyi
- 🎓 2020.9-2023.6 : Shanghai University / **Mphil** / Supervisor : Prof. Zhang Tongyi
- 🎓 2016.9-2020.6 : Beijing University of Chemical Technology / **Bachelor**

Member of Chinese Crystallographic Society (CCrS) ;

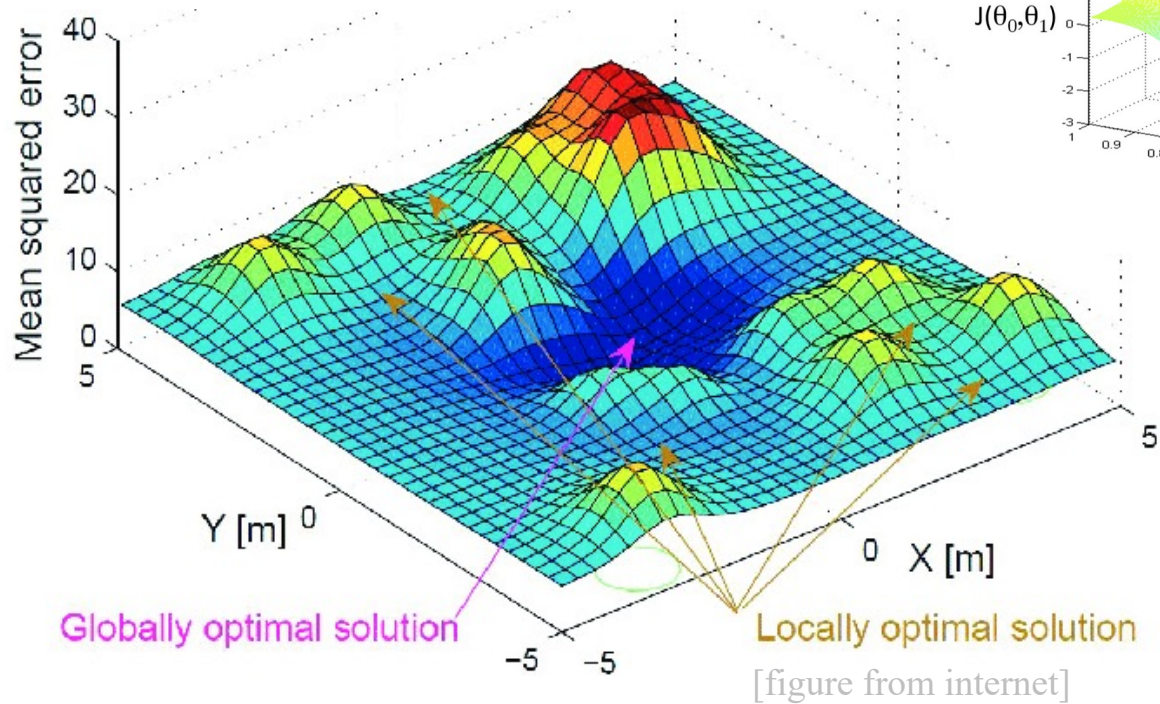
working on ML for crystal structure analysis, using X-ray spectral analysis technology and machine learning techniques. ML-based X-ray technology (XRD、XAFS、XPS) [https://orcid.org/0000-0001-](https://orcid.org/0000-0001-7273-6779)

[7273-6779](https://orcid.org/0000-0001-7273-6779)

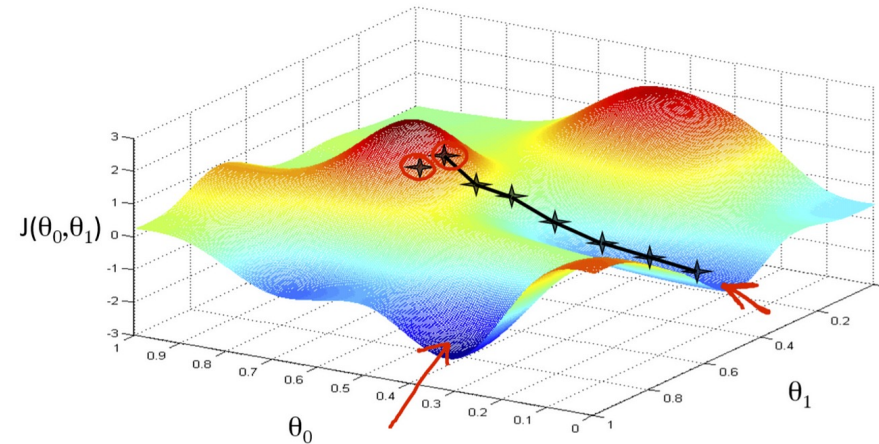


Non-convex Optimization

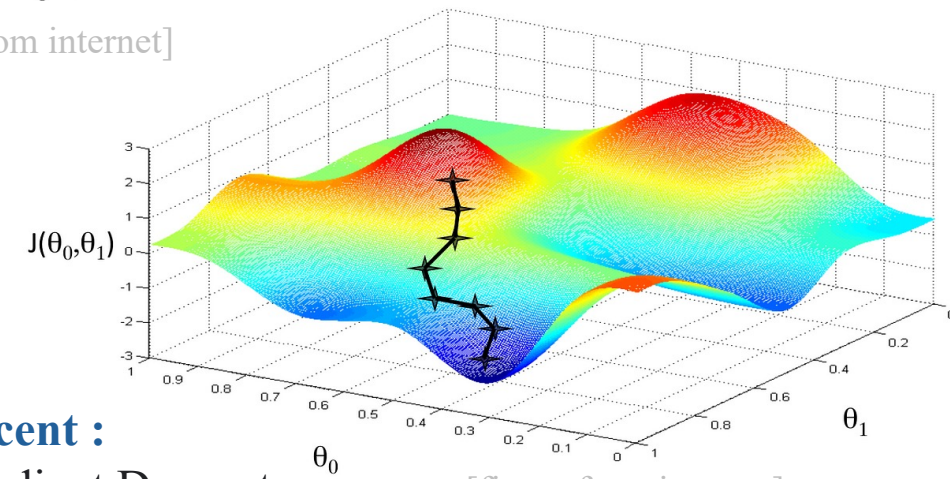
A **non-convex optimization** problem is characterized by having a non-convex objective function or non-convex constraints.



The function is **expressed analytically**



[figure from internet]



[figure from internet]

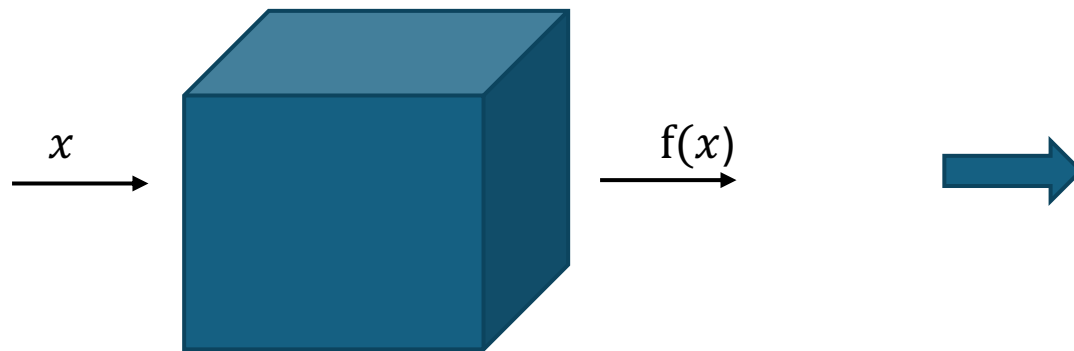
Gradient descent :
Stochastic Gradient Descent
Steepest Gradient Descent
Newton Gradient Descent ...



Non-convex Optimization

Goal : solve $\operatorname{argmin}_x f(x)$

Where $f(x)$ is a black box and is a **time-consuming-to-evaluate** function



Diffacults :

The analytical expression of the function is **not known**

The computation of $f(x)$ is associated with **significant time cost**

Heuristic optimization algorithm:

- particle swarm algorithm
- simulated annealing algorithm
- genetic algorithm

...

Bayesian optimization:

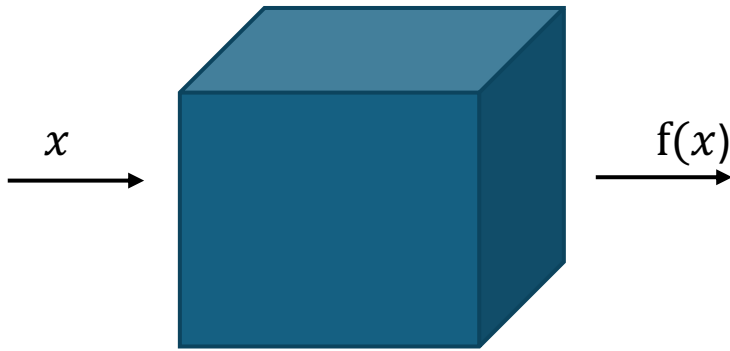
- Expected Improvement algorithm
- Upper confidence bound
- Predictive Entropy Search
- Knowledge Gradient

...



Target of Optimization

The optimization algorithms applied to black-box problems are collectively geared towards solving the **challenge of effective sampling**



$$\begin{aligned}x_1 &\rightarrow f(x_1) \\x_2 &\rightarrow f(x_2) \\&\dots \\x_n &\rightarrow f(x_n) \\&\text{(n is a finite number)} \\x_* &\rightarrow f(x_*)\end{aligned}$$

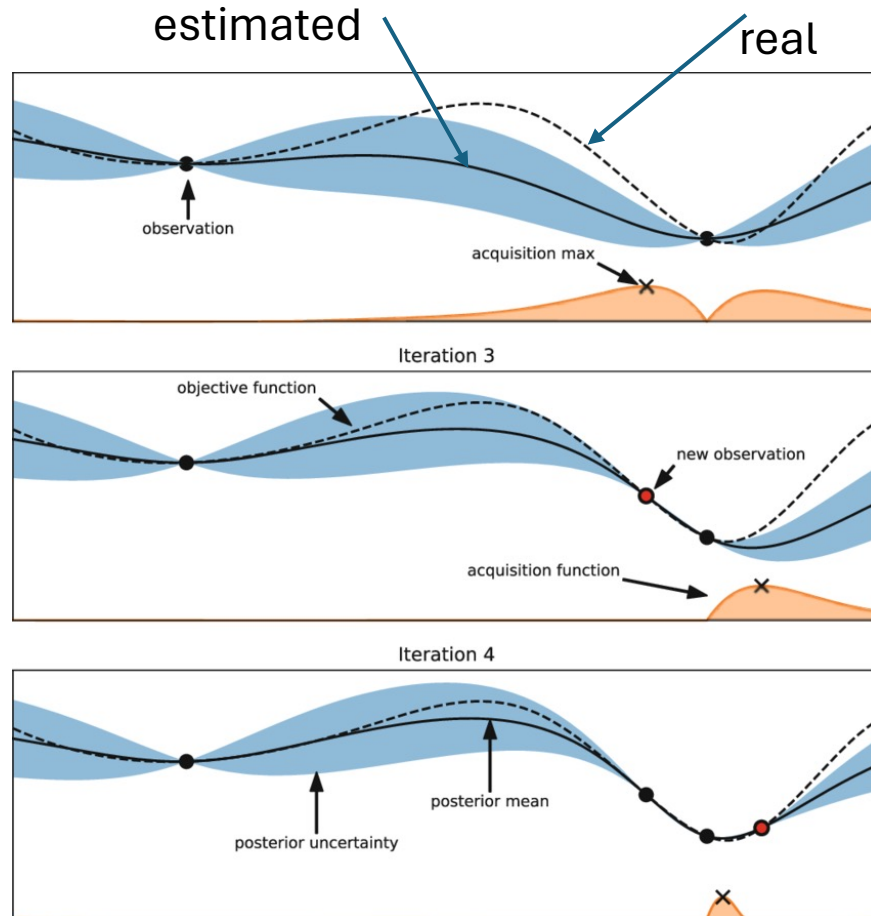
Many optimization algorithms leverage historical information to guide the selection of the next sampling point, determining which infill data,

$x_{n+1} \rightarrow f(x_{n+1})$ is most informative in the pursuit of deriving the optimal solution x_* .



The differences

All optimization algorithms incorporate a fundamental concept of an **optimization (response) surface**



[figure from internet]

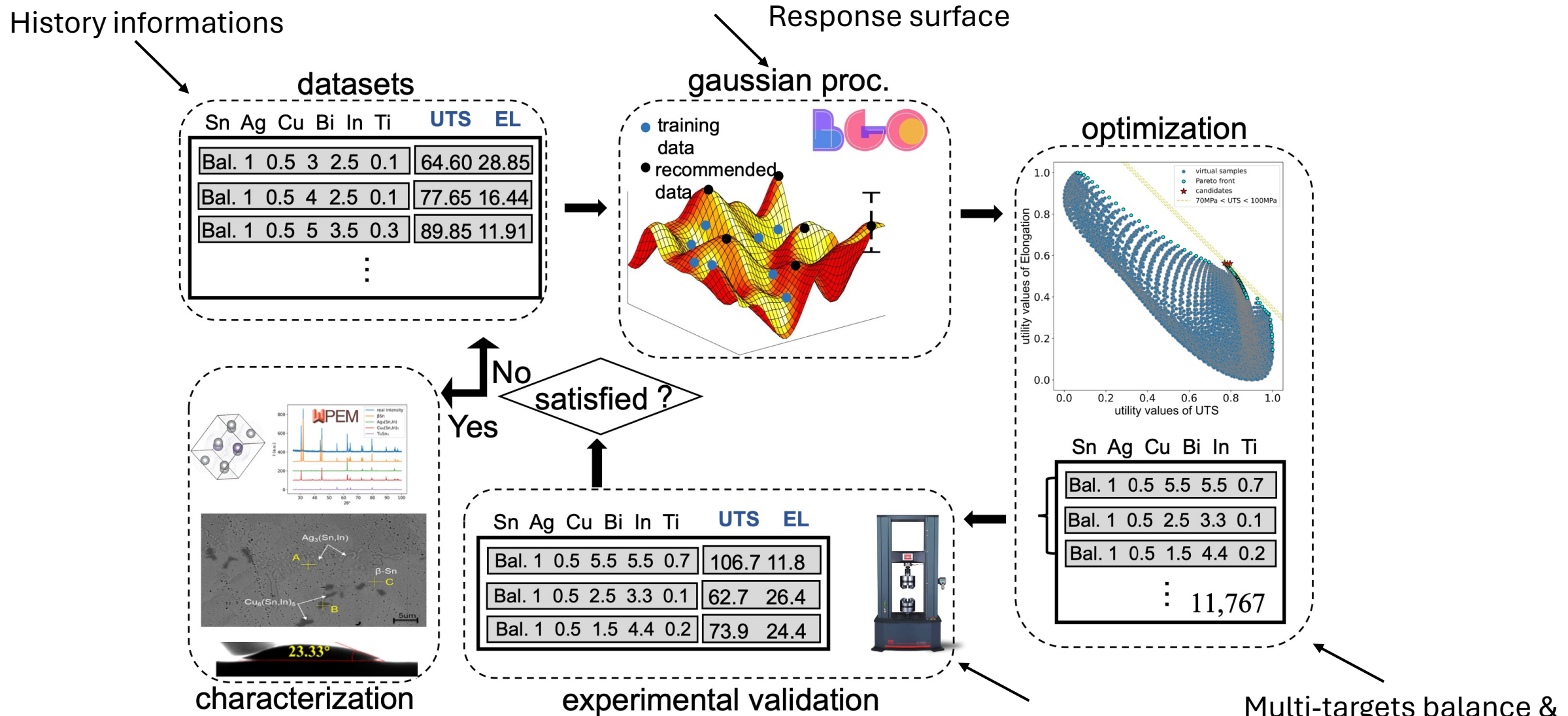
The optimization surface **undergoes changes** with the introduction of infill data.

This response surface can be **implicit**, as seen in heuristic optimization, or **explicit**, as **observed** in Bayesian optimization.

A notable distinction lies in Bayesian optimization's approach, as **it does not unconditionally trust the derived response surface**. Instead, it carefully navigates the **exploration-exploitation trade-off**, balancing the need for exploring new regions with exploiting the current knowledge to achieve optimal results.



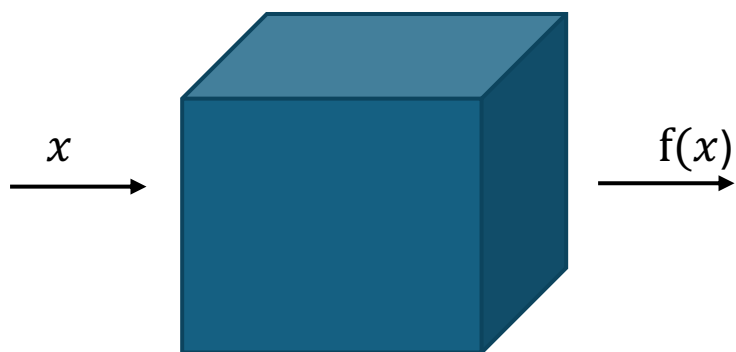
Bayesian global Optimization





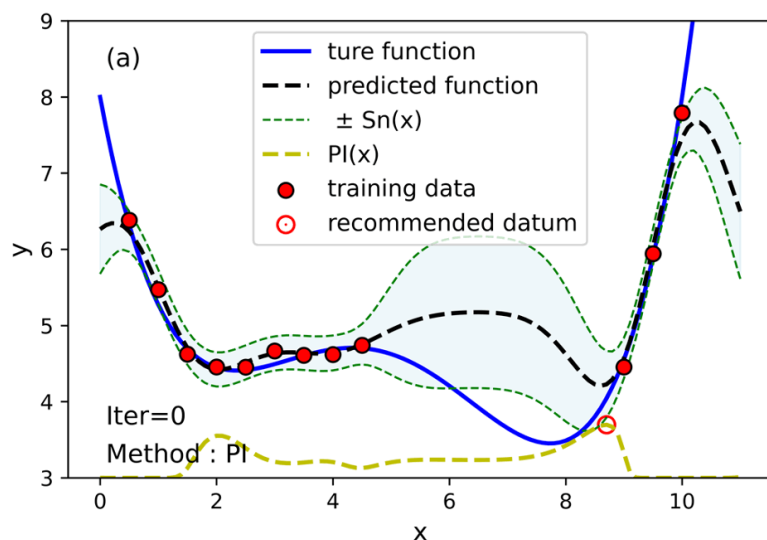
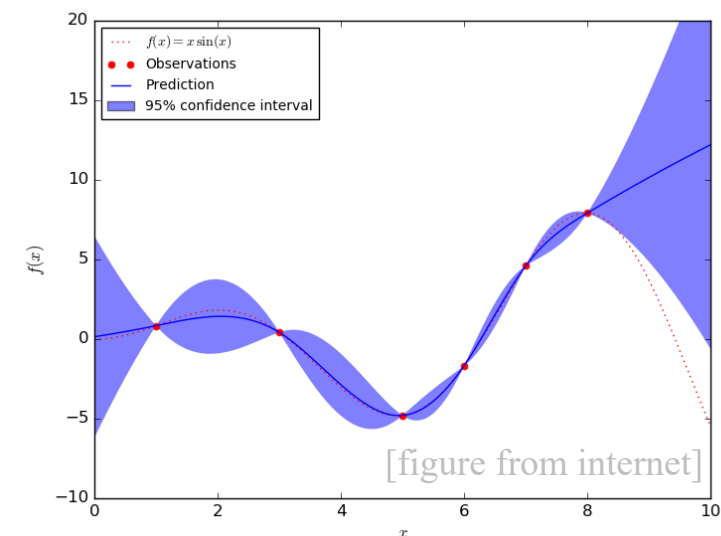
Bayesian global Optimization : Response surface

The first and most important step is to **establish the response surface** with collect data pairs



$$\begin{aligned}x_1 &\rightarrow f(x_1) \\x_2 &\rightarrow f(x_2) \\&\dots \\x_n &\rightarrow f(x_n)\end{aligned}$$

A ML model



Iterations	x_i	$y_i(x_i)$
1-th	8.7000	4.0418

The reliability is insufficient
Bgolearn provided **three modules**
for establishing the surface :

1. Gaussian Process-Based
2. Boosting Sampling-Based (SVM, RF, MLP, etc.)
3. Multi-model-Based



Bayesian global Optimization : Virtual space

In BGO, the **virtual space** refers to the area constituted by all **unevaluated data** at the current step. In a discrete scenario, we are discussing situations where the data points are **distinct**.

In Mathematic :

$$\begin{aligned} x_1 &\rightarrow f(x_1) \\ x_2 &\rightarrow f(x_2) \\ &\dots \\ x_n &\rightarrow f(x_n) \end{aligned}$$

Data be recorded

$$V : x_i (i \notin (1, 2, \dots, n))$$

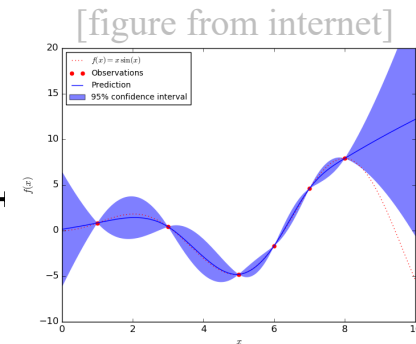
The data within the virtual space, denoted as V , must encompass the optimal solution

$$\begin{aligned} x_1 &\rightarrow f(x_1) \\ x_2 &\rightarrow f(x_2) \\ &\dots \\ x_n &\rightarrow f(x_n) \end{aligned}$$

ML mdoel

Is x_i with the minimum value of $g(x_i)$ considered the best one? In BGO, the model is not entirely convinced, **highlighting the exploration-exploitation tradeoff**

All data within the virtual space are assessed by the response surface, **taking into account associated uncertainties**



$$V : x_i (i \notin (1, 2, \dots, n))$$

$$G : g(x_i) (i \notin (1, 2, \dots, n))$$



Bayesian global Optimization : Ranking

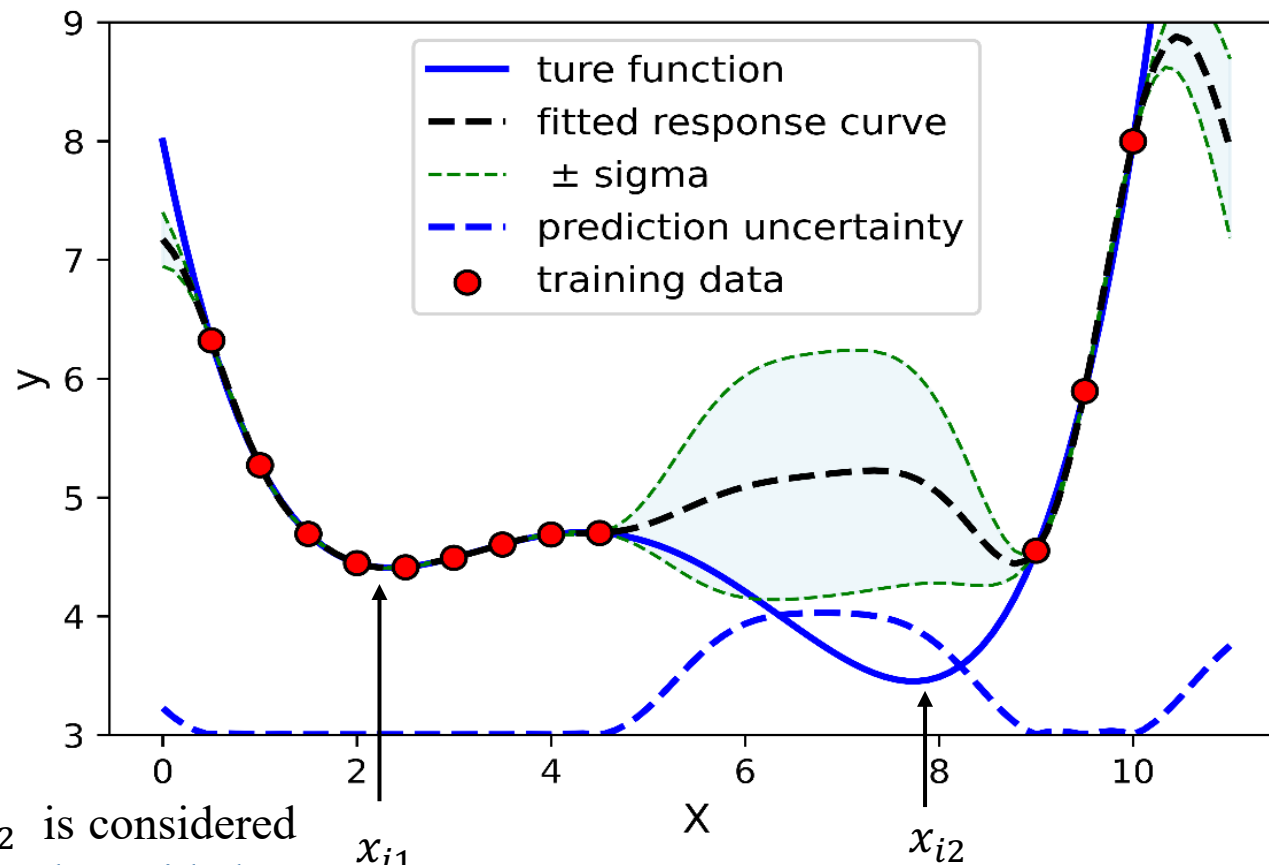
Through the computation of the response surface, we can obtain all the evaluated function values and **their associated uncertainties** for each data point in V

$$V : x_i (i \notin (1, 2, \dots, n))$$

$$G : g(x_i) (i \notin (1, 2, \dots, n))$$

$$S : \text{std}(x_i) (i \notin (1, 2, \dots, n))$$

We use 'g' to denote the evaluated **means** and 'std' to represent the evaluated **standard variance**.



While x_{i1} exhibits the lowest $g(x_{i1})$, however x_{i2} is considered optimal from a global perspective. **How can BGO be guided to recommend solutions closer to x_{i2} ?**



Exploration and Exploitation

How China discovered new oil fields ?



[figure from internet]

In regions with **existing oil fields (A sites)**, there is a heightened likelihood of discovering new oil fields. Conversely, provinces lacking oil fields (**B sites**) exhibit the potential for discovering larger reserves of oil.

Prioritizing attention on A sites **may diminish the probability of finding larger reserves**. Meanwhile, focusing efforts on B sites **may seem futile**, but there have a probability of discovering larger reserves. **The challenge lies in the strategic choice between these options.**



风浪越大，鱼越贵

[figure from internet]



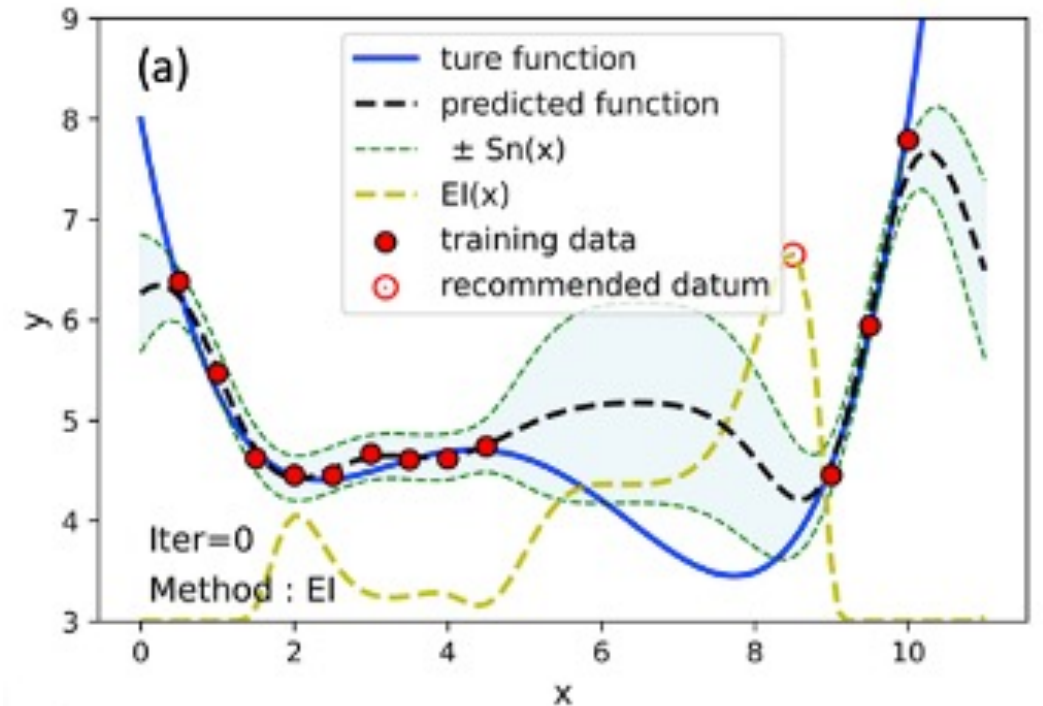
Exploration and Exploitation

The **utility function** is incorporated into BGO, exemplified by the Expectation Maximization algorithm

$$I_n(\mathbf{x}) = \max \left(f_{\min}^{(n)} - g(\mathbf{x}), 0 \right)$$

$$EI_n(\mathbf{x}) = E_{\hat{y}_n(\mathbf{x})} \left(\max \left(f_{\min}^{(n)} - \hat{g}_n(\mathbf{x}), 0 \right) \right)$$

$$EI_n(\mathbf{x}) = s_n \left[\left(\frac{f_{\min}^{(n)} - g_n}{s_n} \right) \Phi \left(\frac{f_{\min}^{(n)} - g_n}{s_n} \right) + \phi \left(\frac{f_{\min}^{(n)} - g_n}{s_n} \right) \right]$$

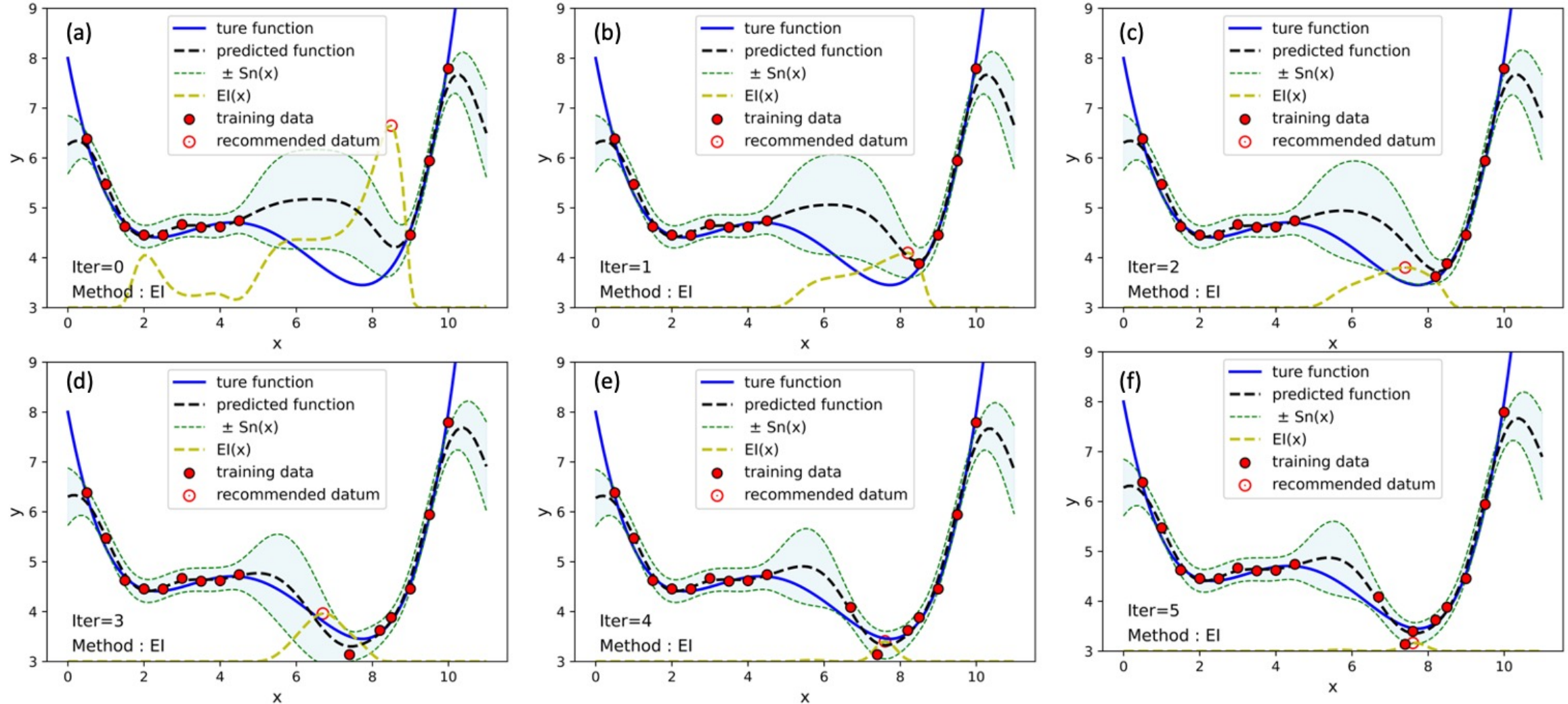


Recommendations are made **by considering both the evaluated values and their associated uncertainties**. The recommended data must then undergo measurement to acquire the real function value (experiments or simulations)



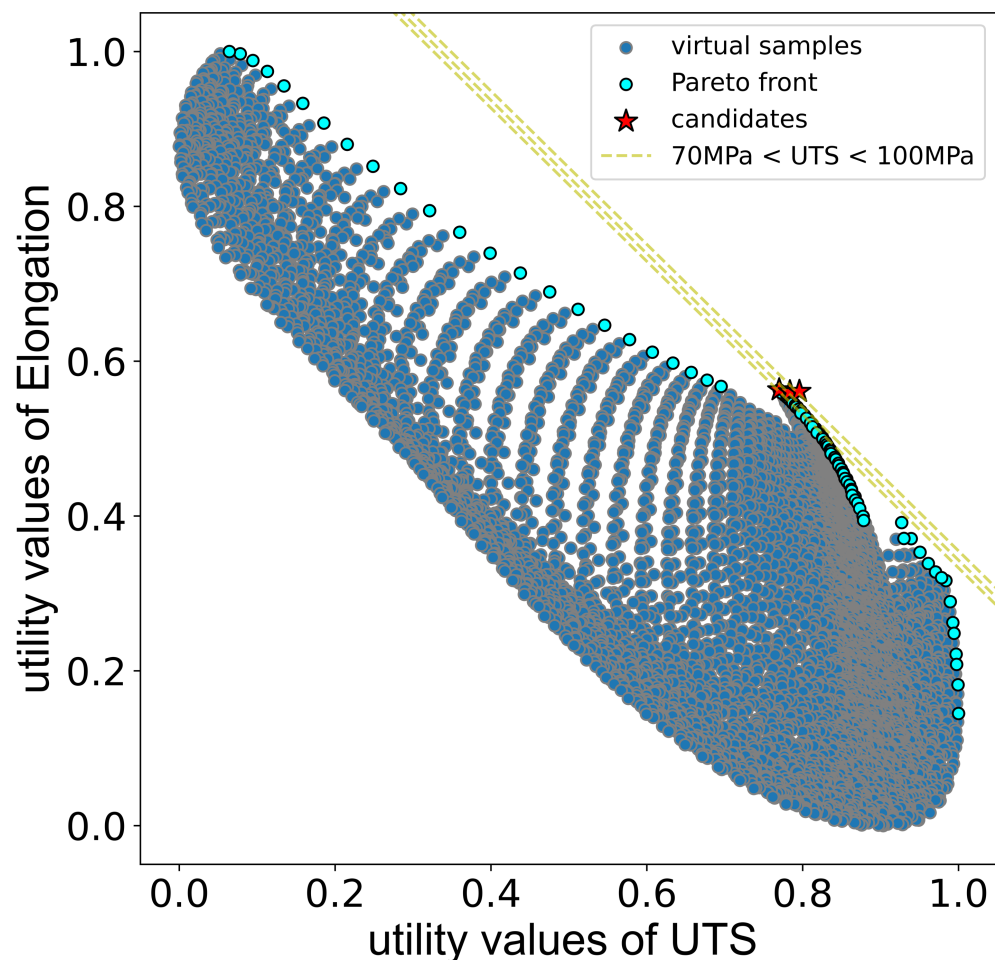
Exploration and Exploitation

The response surface undergoes updates with each iteration





Multi-targets selection :



In the context of multi-tasking, a straightforward approach involves evaluating each data point in the virtual space twice. This yields the 'score' for each data, and a **Pareto space** is then formed.

Selecting a data point from the Pareto front involves various methods:

1. Diagonal search
2. Distance search
3. Equal value search
4. Crowding distance search
5. ...

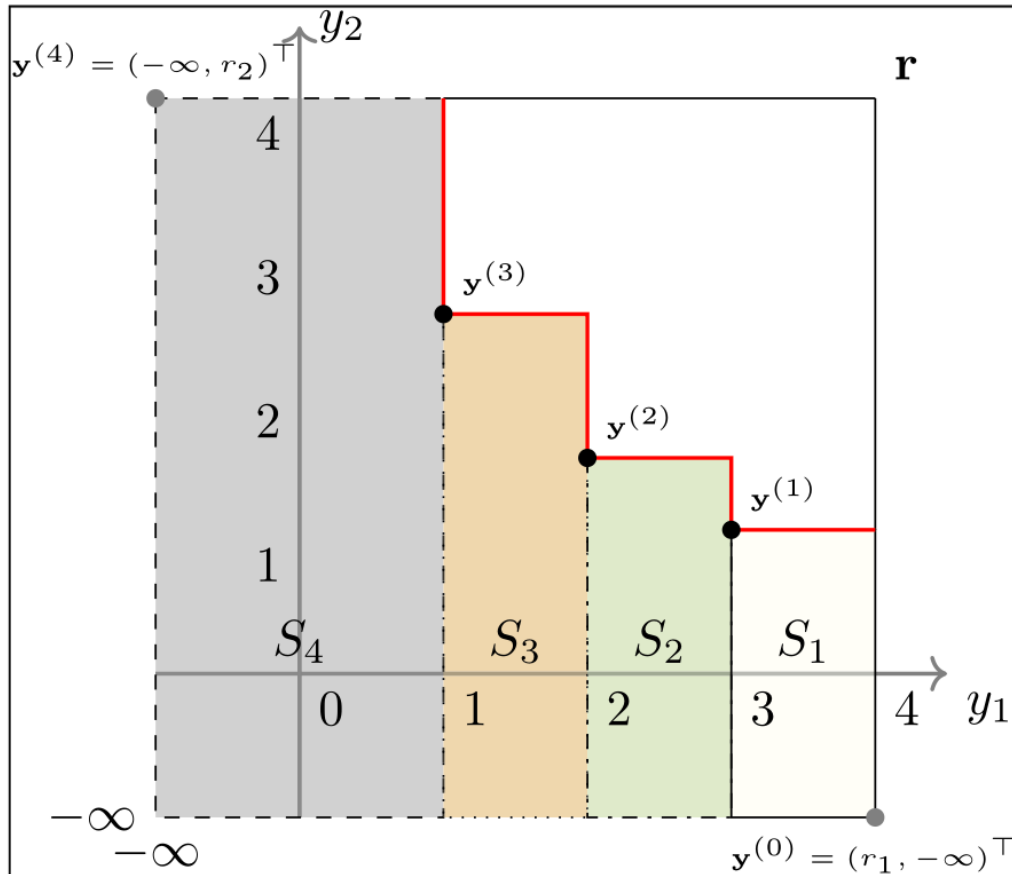
Each method **recommends points located on the front**, contributing to the multi-task optimization process.



Multi-targets BGO :

In multi-target BGO, the evaluation of infill data focuses on enhancing the influence **across all targets**.

Hypervolume Improvement



Algorithm 1 MOBGO algorithm.

Input: Objective functions $\mathbf{y}$, initialization size μ , termination criterion T_c

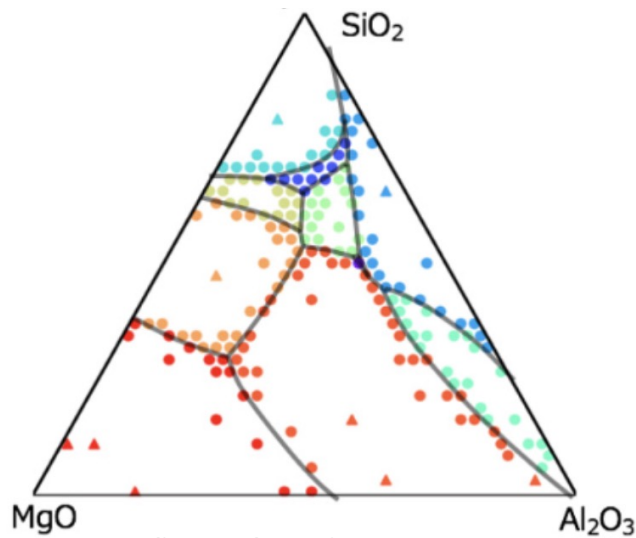
Output: Pareto-front approximation $\mathcal{P}$

- 1 Initialize μ points $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(\mu)})$;
- 2 Evaluate the initial set of μ points: $(\mathbf{y}^{(1)} = \mathbf{y}(\mathbf{x}^{(1)}), \dots, \mathbf{y}^{(\mu)} = \mathbf{y}(\mathbf{x}^{(\mu)}))$;
- 3 Store $(\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(\mu)})$ and $(\mathbf{y}^{(1)} = \mathbf{y}(\mathbf{x}^{(1)}), \dots, \mathbf{y}^{(\mu)} = \mathbf{y}(\mathbf{x}^{(\mu)}))$ in D :
 $D = ((\mathbf{x}^{(1)}, \mathbf{y}^{(1)}), \dots, (\mathbf{x}^{(\mu)}, \mathbf{y}^{(\mu)}))$;
- 4 Compute the non-dominated subset of D and store it in $\mathcal{P}$;
- 5 $g = \mu$;
- 6 **while** $g \leq T_c$ **do**
 - 7 Train Kriging models $M_1, \dots, M_d$ based on D ;
 - 8 Use an optimizer (*opt*) to find the promising point $\mathbf{x}^*$ based on surrogate models M , with the infill criterion C ;
 - 9 Update D : $D = D \cup (\mathbf{x}^*, \mathbf{y}(\mathbf{x}^*))$;
 - 10 Update $\mathcal{P}$ as the non-dominated subset of D ;
 - 11 $g = g + 1$;
- 12 **Return** $\mathcal{P}$



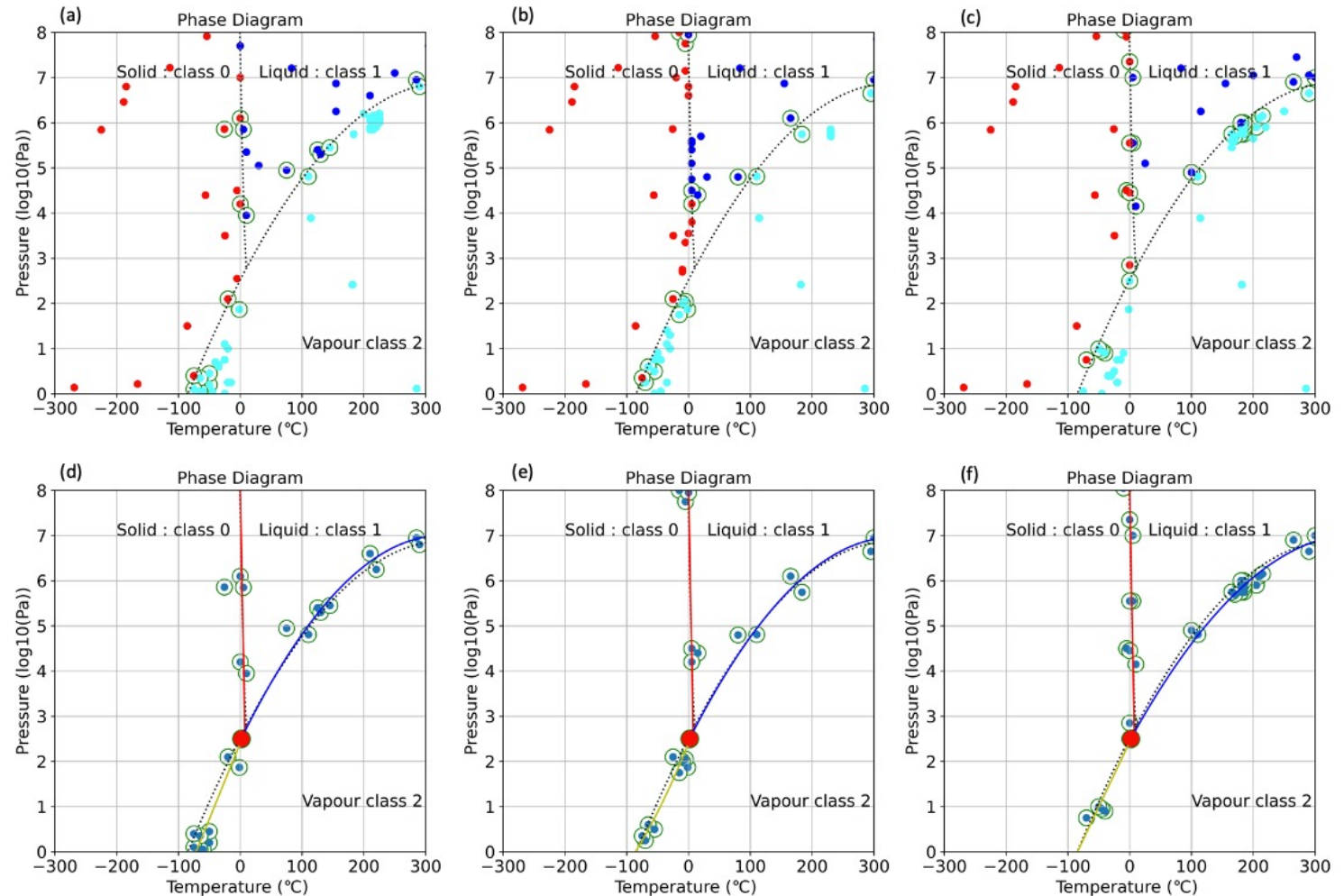
BGO classification

For **classification** problems, BGO proves to be an excellent optimization method in identifying boundaries between different categories



[figure from internet]

Find the phase boundaries





Bgolearn is a computational approach that enables material composition-oriented design and performance-oriented optimization. By leveraging existing experimental data, Bgolearn searches for the optimal material composition design within a specified composition space in order to maximize or minimize the desired performance metric.

Only takes three steps

step 1 : `pip install Bgolearn` 

step 2 :

```
# import your dataset (Samples have been characterized)
# 导入研究的数据集(已经表征过的样本)
data = pd.read_csv('data.csv')
x = data.iloc[:, :-1]
y = data.iloc[:, -1]
# virtual samples which have same feature dimension
with x
# 设计的虚拟样本, 与x具有相同的维度
vs = pd.read_csv('VS.csv')
```

step 3 :

```
# instantiate class
# 实例化类
Bgolearn Bgolearn = BGOS.Bgolearn()
# Pass parameters to the function
# 传入参数
Mymodel = Bgolearn.fit(data_matrix = x,
Measured_response = y, virtual_samples = vs)
# derive the result by EI
# 通过EI导出结果
Mymodel.EI()
```



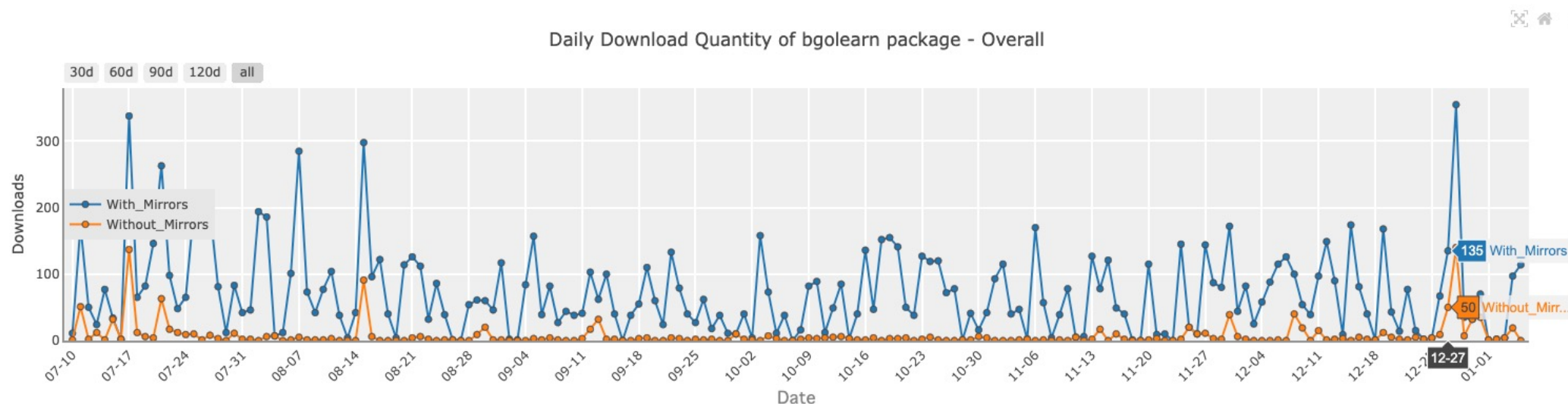
for regression

- 1.Expected Improvement algorithm (期望提升函数)
- 2.Expected improvement with “plugin” (有 “plugin”的期望提升函数)
- 3.Augmented Expected Improvement (增广期望提升函数)
- 4.Expected Quantile Improvement (期望分位提升函数)
- 5.Reinterpolation Expected Improvement (重插值期望提升函数)
- 6.Upper confidence bound (高斯上确界函数)
- 7.Probability of Improvement (概率提升函数)
- 8.Predictive Entropy Search (预测熵搜索函数)
- 9.Knowledge Gradient (知识梯度函数)

for classification

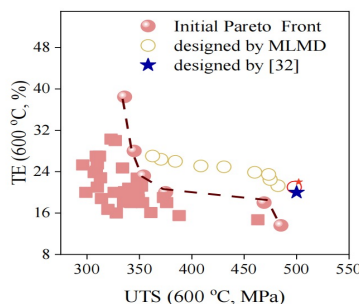
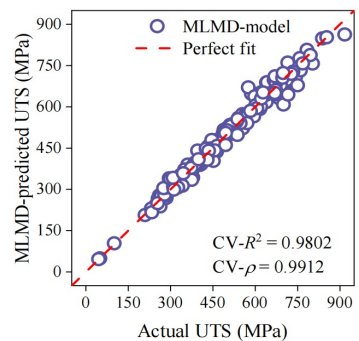
- 1.Least Confidence (欠信度函数)
- 2.Margin Sampling (边界函数)
- 3.Entropy-based approach (熵索函数)

Preserved in various domestic and international mirrors, including Tsinghua University, Alibaba, Tencent, University of Science and Technology of China, etc., with over **5,000 downloads**

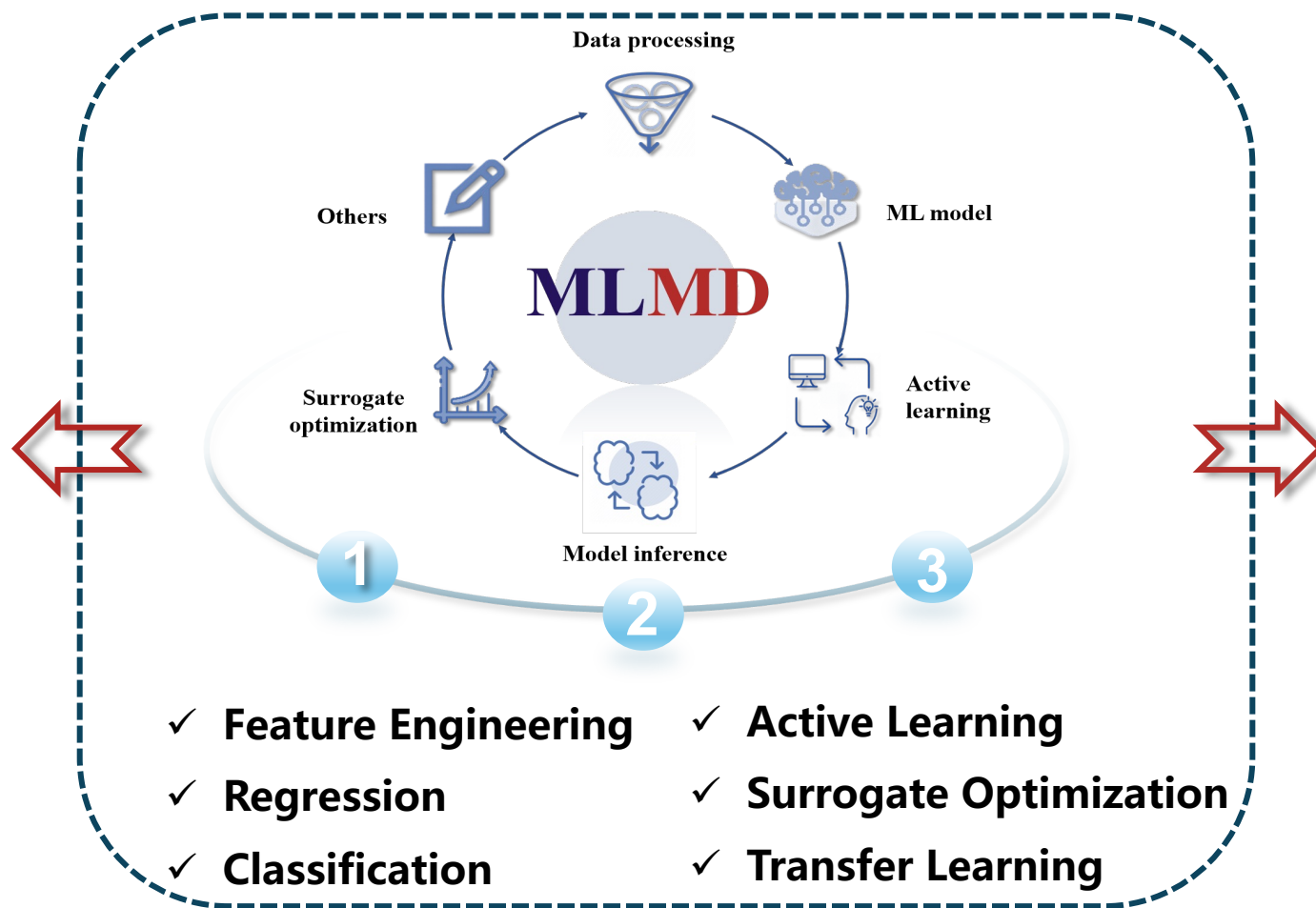




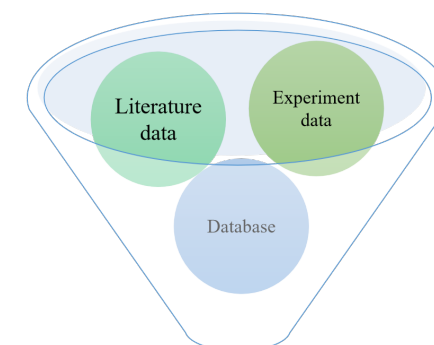
MLMD —Materials design platform



MLMD Prediction	True Class			
	RMG	BMG	CRA	
	RMG	3505	68	179
	BMG	150	1011	6
CRA	405	8	1139	



End-to-end material design



Potential high performance materials

Verified by experiment



MLMD



Code

MLMD (Machine Learning for Material Design) aims at utilizing general and frontier AI algorithms to accelerate the end-to-end material design with programming-free.



Single-objective active learning in MLMD

Single-objective Active Learning

Drag and drop files here
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Browse files

Sn_data_vs.csv 200.1KB

data.csv 1.0KB

1. Upload experiment dataset and visual samples dataset

Data information

rows

1 5 27

	Sn	Bi	In	Ti	Ten	Elong	Tstd	Estd
0	92.9	3	2.5	0.1	64.6	20.85	1.53	0.71
1	92.7	3	2.5	0.3	62.59	23.69	2.59	4.33
2	92.5	3	2.5	0.5	72.05	22.84	0.74	4.99
3	91.9	3	3.5	0.1	66.6	19.79	2.49	2.68
4	91.7	3	3.5	0.3	70.29	21.6	3.41	4.18

Feature and target

target number

2. Split feature variables and target variables

4

	Sn	Bi	In	Ti
0	92.9	3	2.5	0.1
1	92.7	3	2.5	0.3
2	92.5	3	2.5	0.5
3	91.9	3	3.5	0.1
4	91.7	3	3.5	0.3

	Ten	Elong	Tstd	Estd
0	64.6	20.85	1.53	0.71
1	62.59	23.69	2.59	4.33
2	72.05	22.84	0.74	4.99
3	66.6	19.79	2.49	2.68
4	70.29	21.6	3.41	4.18

target

3. Select target variable

target

Ten

Hyper Parameter

mission

Regression

regressor

GaussianProcess

noise std

0.001

4. Set active learning hyperparameters

sample number

1

min search

False

sample criterion

Expected Improvement algorithm

Expected Improvement algorithm

Expected improvement with "plugin"

Augmented Expected Improvement

Expected Quantile Improvement

Reinterpolation Expected Improvement

Upper confidence bound

Probability of Improvement

Predictive Entropy Search

You have uploaded the visual sample point file.

Optimize

5. Start optimization

Train

Recommended Sample

	Sn	Bi	In	Ti
0	87.4	5.5	5.5	0.1

Recommended experiment points

download



Multi-objective active learning in MLMD

Multi-objective Active Learning

Drag and drop files here
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Sn_data_vs.csv 200.1KB

data-multi.csv 0.8KB

1. Upload experiment dataset and visual samples dataset

Data information



	Sn	Bi	In	Ti	Ten	Elong
0	92.9	3	2.5	0.1	64.6	20.85
1	92.7	3	2.5	0.3	62.59	23.69
2	92.5	3	2.5	0.5	72.05	22.84
3	91.9	3	3.5	0.1	66.6	19.79
4	91.7	3	3.5	0.3	70.29	21.6

Feature and target

target number

2

2. The last two columns as target variables

	Sn	Bi	In	Ti
0	92.9	3	2.5	0.1
1	92.7	3	2.5	0.3
2	92.5	3	2.5	0.5
3	91.9	3	3.5	0.1
4	91.7	3	3.5	0.3

	Ten	Elong
0	64.6	20.85
1	62.59	23.69
2	72.05	22.84
3	66.6	19.79
4	70.29	21.6

target

target

Ten × Elong ×

Ten ref location

0

Elong ref location

0

Optimize

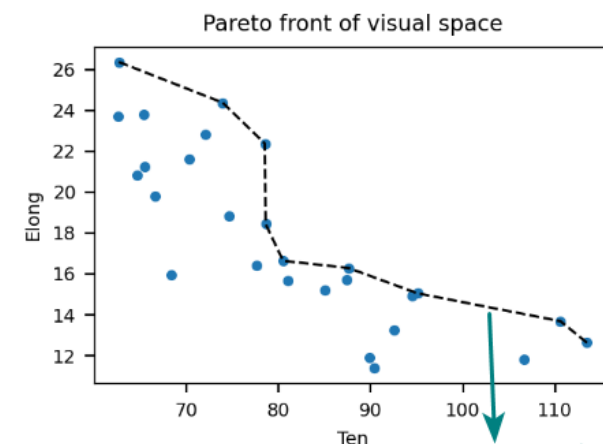
3. Start optimization

Opt

	Sn	Bi	In	Ti
0	93.1	2.4	2.9	0.1

Recommended experiment points

download



Pareto frontier in initial dataset

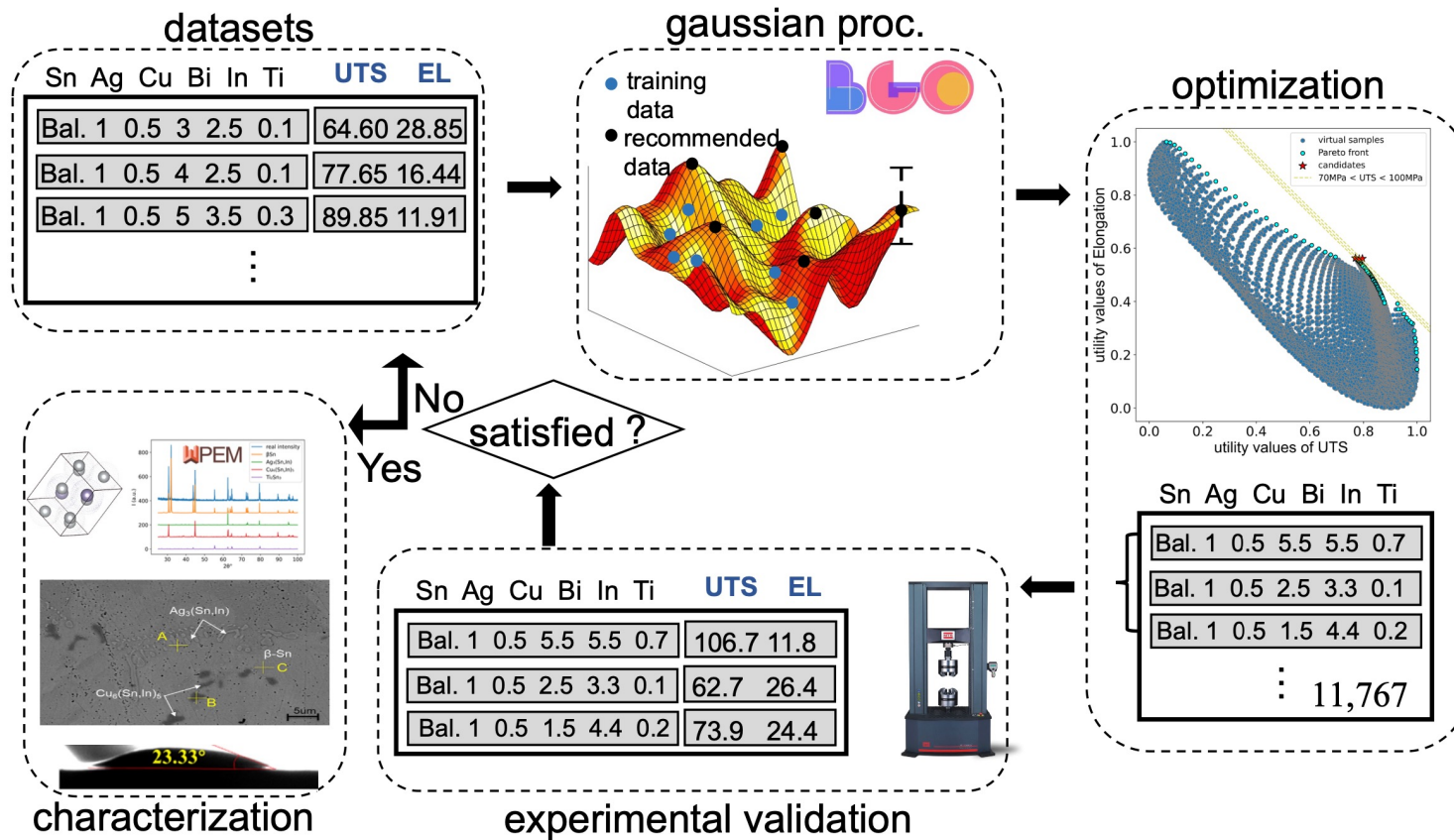
	Ten	Elong
0	113.44	12.64
1	110.56	13.69
2	95.07	15.05
3	87.58	16.29
4	80.49	16.64
5	78.59	18.44
6	78.49	22.38
7	73.94	24.37
8	62.67	26.36

download



Baysian global Optimization

Numerous scientific problems require careful consideration



1. "What types of models and methodologies can be employed to estimate the mean and variance of candidates?
(DeepNN, for instance, excels in high-dimensional spaces.)"

2. "How can one define a reasonable search space (virtual space)?
(A space that is too large may introduce excessive randomness, while one that is too small might result in missing the optimal solutions.)"

3. "How does one strike a balance in the exploration-exploitation trade-off?
(Especially, how should uncertainty be considered in multi-task optimization?)"

...

Cao, Bin and Su, Tianhao ... and Zhang, Tong-Yi. Available at

SSRN: <https://ssrn.com/abstract=4686075> or <http://dx.doi.org/10.2139/ssrn.4686075>



Thank you!

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